

# Four-Level SapBERT: Improvement Study

## Revised comparison | Levels 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent

This improvement study uses the same 10,000 supplied records, frozen SapBERT embeddings and original grouped partitions. It expands the search to PCA-64, PCA-128 and PCA-256, stronger/weaker regularization, class balancing, PCA whitening and structured-only controls. No labels are changed.

### Selected: SapBERT + PCA-128 + Logistic Regression

Training settings: C=10; balanced class weights.

Patient preprocessing includes quadratic numeric terms; the classifier family and frozen SapBERT encoder are unchanged.

An additional 19 explicit detail features preserve severity, onset, duration and other mentions from the original complaint alongside the SapBERT concept embedding.

Selected total input size: 197 features. PCA retained variance: 95.51%.

| Retrospective metric (%) | Initial | Prior round | Selected |
|--------------------------|---------|-------------|----------|
| Accuracy                 | 85.19   | 88.99       | 89.29    |
| Macro precision          | 85.71   | 89.37       | 89.67    |
| Macro recall             | 85.90   | 89.50       | 89.81    |
| Macro F1                 | 85.80   | 89.43       | 89.73    |
| Emergency recall         | 92.16   | 93.87       | 95.10    |
| Under-triage             | 7.60    | 4.65        | 4.65     |

Selected quadratic weighted kappa: 0.9504; mean absolute level error: 0.1081; over-triage: 6.05% (initial 7.20%).

Selected retrospective MCC: 0.8563; log loss: 0.2419; multiclass Brier score: 0.1497.

Mean development CV macro F1 change versus initial: +3.67 percentage points. Selection uses all five development folds and an emergency-recall constraint; the test results above do not select the winner.

Paired development bootstrap versus initial: F1 gain 95% interval +2.97 to +4.33 percentage points. This conditional interval excludes zero. This diagnostic excludes model-selection uncertainty.

Evaluation scope: These 1,999 test rows were already examined in the previous study. The new scores are retrospective comparisons, not a fresh independent estimate. New labelled records are needed to confirm generalization. Saved concepts are evaluated here; live translation accuracy is not measured.

Trade-offs remain visible: compare emergency recall and under-triage as well as aggregate scores. SapBERT is not fine-tuned; these gains do not establish clinical validity.

# Complete Development Cross-Validation

All 68 configurations; 340 fits across five grouped folds. F1, accuracy and emergency recall are percentages. SD is the fold-to-fold F1 standard deviation in percentage points. The original reference is lr\_pca64\_c10\_balanced1.

| Configuration                 | F1    | SD   | Accuracy | Emergency recall |
|-------------------------------|-------|------|----------|------------------|
| lr_detail128_c10              | 89.65 | 0.26 | 89.28    | 94.06            |
| lr_detail64_c100              | 89.56 | 0.82 | 89.19    | 93.93            |
| logreg__complaint_details     | 89.38 | 0.50 | 88.99    | 93.63            |
| lr_detail128_c30              | 89.31 | 0.38 | 88.91    | 93.93            |
| lr_detail128_unbalanced       | 89.30 | 0.33 | 88.89    | 93.01            |
| lr_detail128_c1000            | 89.08 | 0.52 | 88.66    | 93.26            |
| lr_detail128_c300             | 89.00 | 0.57 | 88.58    | 93.44            |
| lr_detail128_c1               | 88.68 | 0.63 | 88.29    | 94.42            |
| lr_detail256_c100             | 88.65 | 0.36 | 88.21    | 93.01            |
| hgb__complaint_details        | 88.07 | 0.95 | 87.65    | 92.89            |
| logreg__concept_details       | 87.83 | 1.06 | 87.40    | 93.38            |
| hgb_detail64_leaf7_I220       | 87.68 | 0.73 | 87.25    | 92.77            |
| hgb_detail64_leaf15_I210      | 86.86 | 0.83 | 86.40    | 91.73            |
| lr_quad128_c100               | 86.57 | 0.70 | 86.09    | 92.65            |
| lr_pca128_c100_balanced0      | 86.57 | 0.59 | 86.05    | 91.60            |
| lr_refine128_c300             | 86.50 | 0.75 | 86.00    | 92.52            |
| lr_pca128_c1000_balanced1     | 86.48 | 0.69 | 85.98    | 92.28            |
| lr_quad128_c300               | 86.48 | 0.68 | 85.99    | 92.34            |
| lr_pca128_c100_balanced1      | 86.46 | 0.58 | 85.98    | 92.65            |
| lr_quadratic64_c100_balanced1 | 86.41 | 0.80 | 85.93    | 92.83            |
| lr_pca128_c10_balanced0       | 86.39 | 0.57 | 85.84    | 91.73            |
| lr_refine128_c30              | 86.32 | 0.51 | 85.81    | 92.89            |
| lr_quadratic64_c100_balanced0 | 86.31 | 0.51 | 85.81    | 91.05            |
| lr_refine64_c30               | 86.26 | 0.61 | 85.75    | 93.50            |
| lr_pca128_c1000_balanced0     | 86.24 | 0.47 | 85.71    | 91.05            |
| lr_pca256_c10_balanced0       | 86.24 | 0.44 | 85.68    | 91.67            |
| lr_quadratic64_c10_balanced0  | 86.23 | 0.66 | 85.71    | 91.24            |
| lr_quadratic64_c10_balanced1  | 86.21 | 0.52 | 85.71    | 93.08            |
| lr_pca64_c1000_balanced0      | 86.19 | 0.54 | 85.68    | 90.93            |
| lr_pca64_c100_balanced1       | 86.19 | 0.46 | 85.69    | 93.01            |
| lr_quad256_c100               | 86.18 | 0.92 | 85.66    | 91.54            |
| lr_pca64_c100_balanced0       | 86.17 | 0.78 | 85.65    | 91.05            |
| lr_pca128_c10_balanced1       | 86.16 | 0.43 | 85.63    | 93.07            |
| lr_refine64_c300              | 86.06 | 0.52 | 85.55    | 92.83            |
| lr_pca256_c100_balanced0      | 86.06 | 0.57 | 85.50    | 90.75            |

# Development Cross-Validation: Continued

| Configuration               | F1    | SD   | Accuracy | Emergency recall |
|-----------------------------|-------|------|----------|------------------|
| lr_pca256_c10_balanced1     | 86.05 | 0.46 | 85.51    | 92.95            |
| lr_pca64_c1000_balanced1    | 86.04 | 0.55 | 85.53    | 92.77            |
| hgb_concept_details         | 86.02 | 1.03 | 85.54    | 91.48            |
| lr_pca64_c10_balanced1      | 85.98 | 0.53 | 85.46    | 93.38            |
| lr_pca256_c100_balanced1    | 85.91 | 0.70 | 85.36    | 91.97            |
| lr_quad256_c300             | 85.90 | 1.07 | 85.35    | 91.24            |
| lr_pca64_c10_balanced0      | 85.86 | 0.61 | 85.31    | 91.48            |
| hgb_detail64_leaf31_l220    | 85.85 | 0.45 | 85.40    | 91.12            |
| lr_pca256_c1000_balanced1   | 85.14 | 0.76 | 84.54    | 90.99            |
| lr_pca256_c1000_balanced0   | 85.12 | 0.67 | 84.50    | 90.32            |
| lr_pca128_whiten            | 85.06 | 0.46 | 84.51    | 92.34            |
| lr_pca64_whiten             | 85.04 | 0.57 | 84.54    | 92.09            |
| hgb_refine64_leaf7          | 84.76 | 0.80 | 84.26    | 90.38            |
| hgb_fused_64_leaf7          | 84.34 | 0.65 | 83.81    | 91.36            |
| hgb_fused_64_leaf15         | 84.24 | 0.76 | 83.74    | 90.81            |
| hgb_fused_128_leaf15        | 83.75 | 0.59 | 83.23    | 90.44            |
| lr_pca256_whiten            | 83.49 | 0.82 | 82.89    | 90.56            |
| hgb_refine64_leaf31         | 83.45 | 0.63 | 82.94    | 90.14            |
| hgb_fused_128_leaf7         | 83.41 | 0.98 | 82.83    | 91.61            |
| lr_quadratic64_c1_balanced1 | 83.21 | 0.55 | 82.64    | 93.02            |
| lr_quadratic64_c1_balanced0 | 83.20 | 0.73 | 82.56    | 90.26            |
| rf_detail64_fraction0.3     | 81.96 | 0.63 | 81.58    | 91.79            |
| rf_detail64_fraction0.7     | 81.55 | 0.88 | 81.10    | 88.73            |
| rf_complaint_details        | 80.32 | 0.79 | 80.24    | 95.89            |
| rf_concept_details          | 80.00 | 0.57 | 79.93    | 96.32            |
| rf_refine64_leaf1           | 79.21 | 0.34 | 79.13    | 95.96            |
| rf_refine64_leaf3           | 78.31 | 0.42 | 78.33    | 96.63            |
| rf_fused_64                 | 78.31 | 1.21 | 77.82    | 87.68            |
| rf_refine128_leaf3          | 76.48 | 0.76 | 76.72    | 96.63            |
| rf_refine128_leaf1          | 76.48 | 0.85 | 76.64    | 95.77            |
| hgb_structured_0_leaf7      | 76.23 | 1.18 | 75.55    | 87.38            |
| hgb_structured_0_leaf15     | 75.87 | 1.00 | 75.15    | 86.40            |
| rf_structured_0             | 75.50 | 0.73 | 74.85    | 86.03            |

Selection: highest mean macro F1 among candidates whose mean emergency recall is within one percentage point of the original reference; accuracy resolves ties. A higher F1 does not qualify a candidate whose emergency recall falls below the threshold. This is not a clinical safety guarantee.

# Original 768-D versus PCA-64 Baselines

All original fixed comparisons are retained below. They use the same 1,999 previously examined test rows. Values are percentages; precision, recall and F1 are macro-averaged. These baseline runs are from the first round and are not new experiments.

## Text only

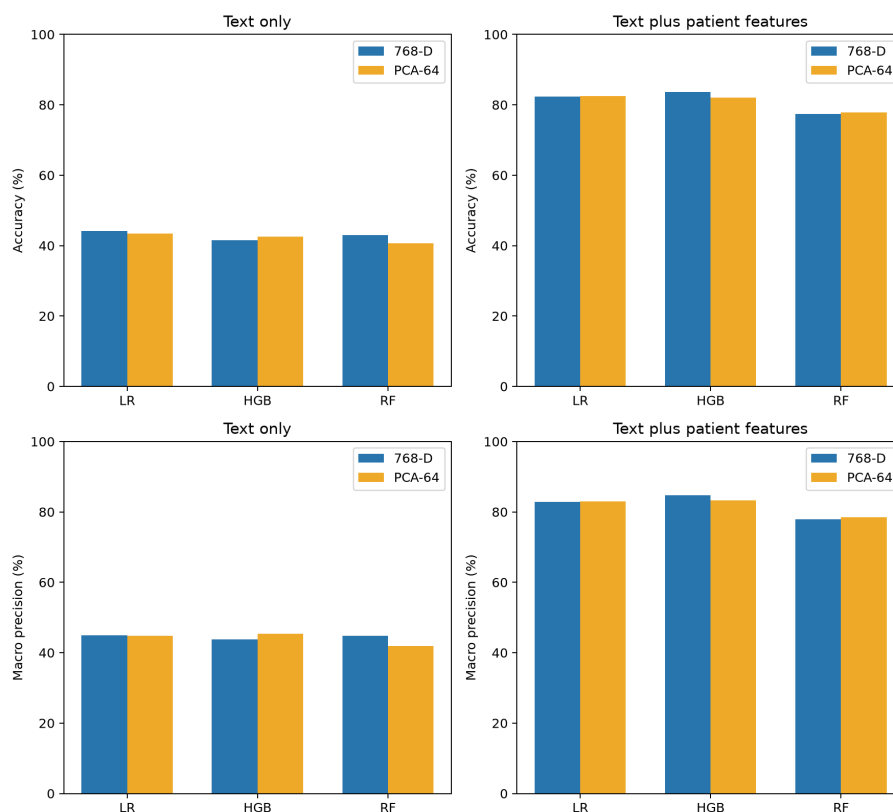
| Text size | Classifier           | Accuracy | Precision | Recall | F1    |
|-----------|----------------------|----------|-----------|--------|-------|
| 768       | Logistic Regression  | 44.17    | 44.89     | 44.87  | 43.95 |
| 768       | HistGradientBoosting | 41.52    | 43.72     | 40.95  | 41.70 |
| 768       | Random Forest        | 42.92    | 44.82     | 42.93  | 43.49 |
| 64        | Logistic Regression  | 43.42    | 44.74     | 44.20  | 43.26 |
| 64        | HistGradientBoosting | 42.57    | 45.34     | 41.85  | 42.80 |
| 64        | Random Forest        | 40.62    | 41.88     | 40.93  | 41.08 |

## Text plus patient features

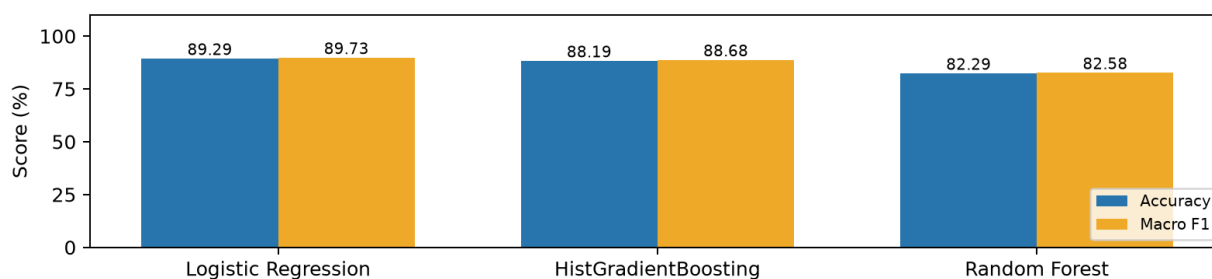
| Text size | Classifier           | Accuracy | Precision | Recall | F1    |
|-----------|----------------------|----------|-----------|--------|-------|
| 768       | Logistic Regression  | 82.29    | 82.89     | 83.10  | 82.92 |
| 768       | HistGradientBoosting | 83.64    | 84.72     | 84.03  | 84.21 |
| 768       | Random Forest        | 77.44    | 77.87     | 78.24  | 77.93 |
| 64        | Logistic Regression  | 82.44    | 83.01     | 83.22  | 83.03 |
| 64        | HistGradientBoosting | 82.09    | 83.28     | 82.54  | 82.76 |
| 64        | Random Forest        | 77.89    | 78.47     | 78.65  | 78.35 |

The original fused PCA-64 input has 86 features: 64 text components and 22 encoded patient features. The full fused input has 790 features. All training transformations are fitted without validation/test rows.

# Performance Graphs



## Tuned classifiers: retrospective comparison

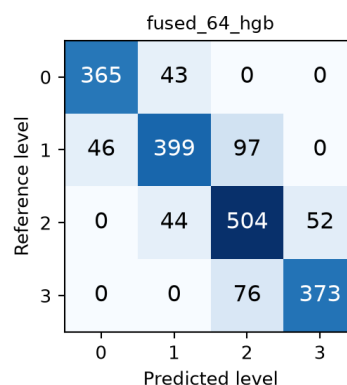
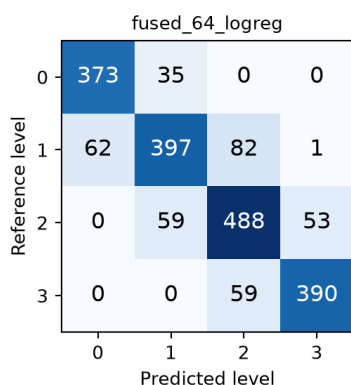
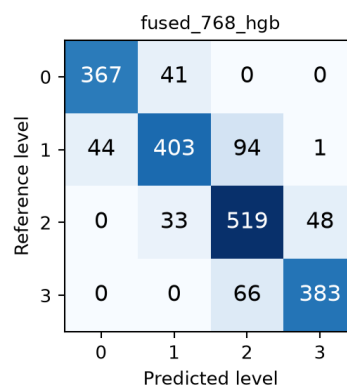
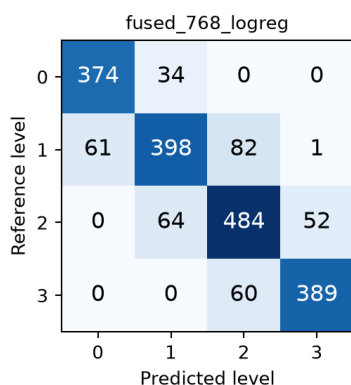
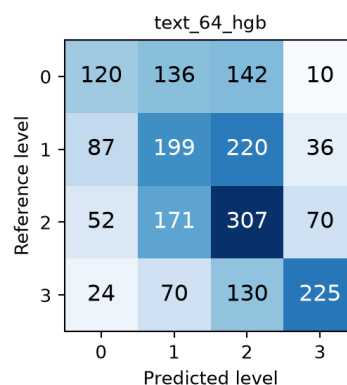
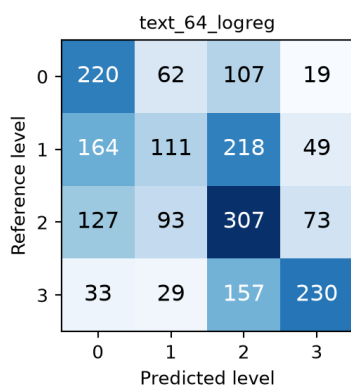
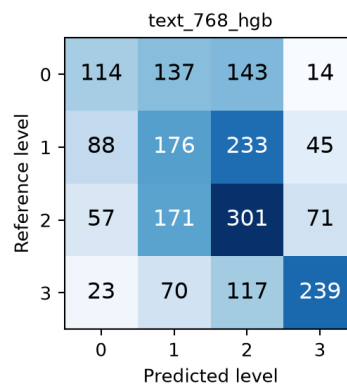
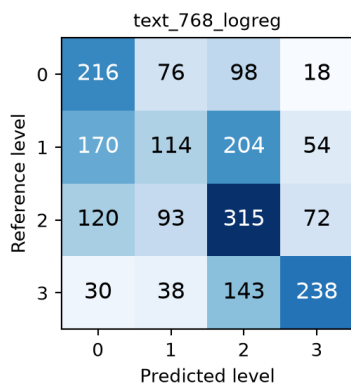


| Classifier           | Accuracy | Precision | Recall | F1    |
|----------------------|----------|-----------|--------|-------|
| Logistic Regression  | 89.29    | 89.67     | 89.81  | 89.73 |
| HistGradientBoosting | 88.19    | 88.86     | 88.58  | 88.68 |
| Random Forest        | 82.29    | 83.00     | 82.76  | 82.58 |

Each family uses its highest fused development CV F1 setting. These descriptive choices do not override the emergency-recall constraint for deployment. Scores are percentages; precision, recall and F1 are macro averages.

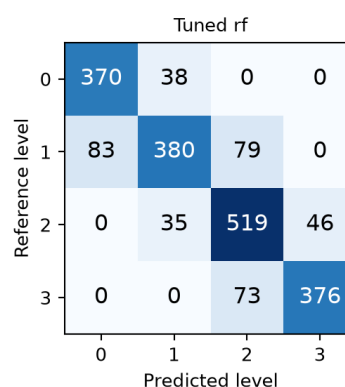
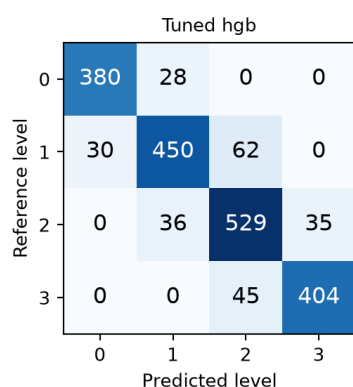
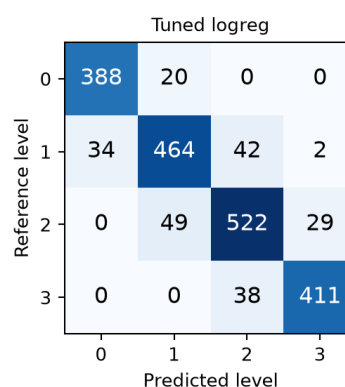
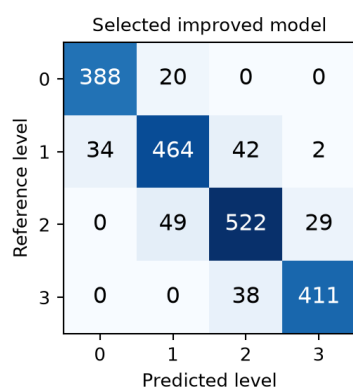
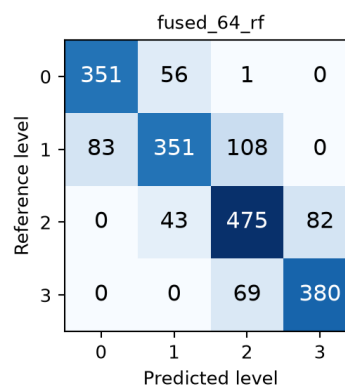
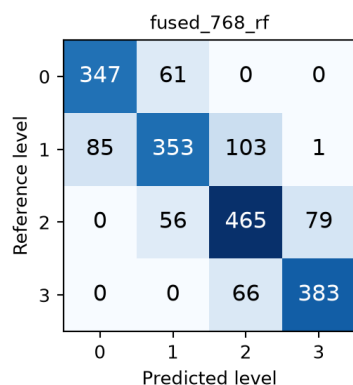
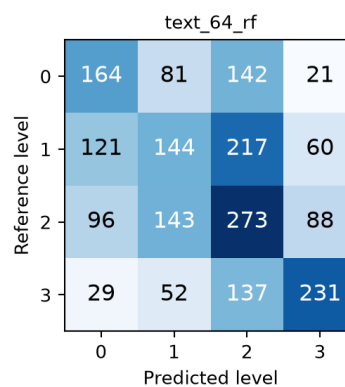
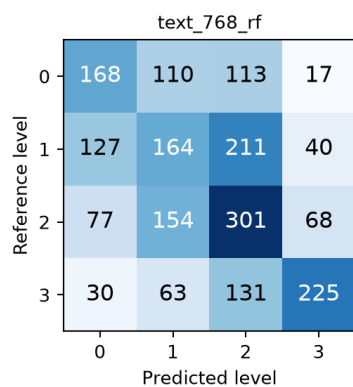
Blue and yellow bars preserve the original comparison style. The selected improved result is shown separately on page 1 because it follows a larger development search.

# Baseline Confusion Matrices: LR and HGB



Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

# Baseline and Selected Confusion Matrices



Rows are reference labels; columns are predictions. Level order: 0 Emergency, 1 Urgent, 2 Standard, 3 Non-urgent. Baseline matrices are unchanged from round one.

## Class Results and Error Review

| Level        | Precision (%) | Recall (%) | F1 (%) | Support |
|--------------|---------------|------------|--------|---------|
| 0 Emergency  | 91.94         | 95.10      | 93.49  | 408     |
| 1 Urgent     | 87.05         | 85.61      | 86.33  | 542     |
| 2 Standard   | 86.71         | 87.00      | 86.86  | 600     |
| 3 Non-urgent | 92.99         | 91.54      | 92.26  | 449     |

## Development out-of-fold confusion matrix

| Reference / predicted | 0    | 1    | 2    | 3    |
|-----------------------|------|------|------|------|
| 0                     | 1535 | 97   | 0    | 0    |
| 1                     | 141  | 1851 | 177  | 3    |
| 2                     | 0    | 162  | 2090 | 148  |
| 3                     | 0    | 0    | 130  | 1667 |

An input audit found no identical development input combinations with conflicting targets. This does not establish that labels are clinically correct. A local review CSV lists development errors and model confidence; it is not committed because it contains individual records.

Use the review list to investigate unclear label boundaries, missing symptom severity/duration/history and data entry issues. Do not replace reference labels with model predictions merely to improve agreement.

## Literature context (different task)

| Model                      | Accuracy (%) | Recall (%) | F1 (%) |
|----------------------------|--------------|------------|--------|
| TF-IDF + LR                | 75.00        | 98.80      | 54.40  |
| TF-IDF + RF                | 75.10        | 96.40      | 58.20  |
| BiLSTM                     | 74.60        | 84.60      | 67.00  |
| CNN                        | 72.30        | 78.70      | 66.70  |
| Current four-level SapBERT | 89.29        | 89.81      | 89.73  |

Seo et al., Scientific Reports (2025), Table 2. DOI: 10.1038/s41598-025-99874-0. Different binary KTAS task; publication metric definitions differ. Context only, not evidence of superiority over these models.

# Methods, Reproducibility and Limits

## Data and preprocessing

The original 8,001 development / 1,999 test partition and five grouped folds are reused. Groups do not overlap across training and validation/test boundaries. Input and embedding hashes are checked before and after the run. Median imputation, categorical encoding, scaling and PCA are learned inside each development fold. No label-derived columns are model inputs.

## Encoder and search

SapBERT-from-PubMedBERT-fulltext uses frozen 768-D CLS embeddings, L2 normalization, a 64-token limit and the original pinned revision. Logistic Regression varies  $C=10$ , 100 and 1000 with and without balancing, plus whitened PCA with  $C=0.1$ . Six additional LR configurations use quadratic numeric terms, with  $C=1/10/100$  and optional balancing. HGB tests 7/15 leaves with regularization; Random Forest tests structured and fused inputs. Every candidate is evaluated on all five folds.

The refinement adds 14 configurations: balanced LR  $C=30/300$  at PCA-64/128; quadratic LR  $C=100/300$  at PCA-128/256; HGB with 7/31 leaves and 800/400 iterations; and RF with 500 trees, square-root feature sampling and leaf sizes 1/3. Classifier families remain unchanged.

Six additional settings compare 19 explicit details extracted from English concepts or original complaints, using each family's prior best setting. The separate learning-curve audit uses nested whole-group subsets; labels remain unchanged.

## Selection and deployment

The incumbent is included in the same search. Selection is recorded before the retrospective test evaluation. The exported model must reproduce saved predictions through the shared GUI/CLI preprocessing. Model artefacts and the incumbent are retained separately. Quadratic terms, if selected, are applied by the fitted numeric preprocessing pipeline. No label correction or encoder fine-tuning is claimed.

## Research interpretation

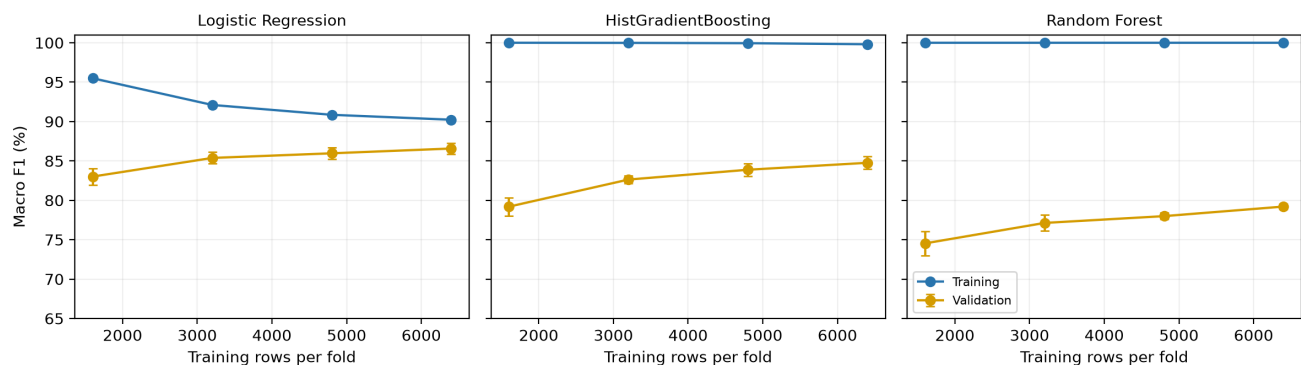
The larger search can overfit development cross-validation. The old test set is already exposed, so its scores must not be presented as untouched confirmation. A new independently labelled dataset is needed for confirmation. Supplied/recovered concepts bypass live translation during this evaluation. The provider could not supply label-assignment rules; independent per-record review is not documented. These are research comparisons, not clinical validation.

## Files

protocol.json freezes candidates and source identities; cross\_validation.csv contains every fit; cv\_summary.csv reports every candidate; selection.json records the decision; retrospective\_results.json contains class scores and matrices. Source data, embeddings and per-record error files remain local.

# Complaint Audit, Learning Curves and Next Steps

The development audit flags 567 explicit duration disagreements between original complaints and supplied concepts, 142 durations not recovered from concepts and 48 family-word omissions. These lexical flags require review; they do not establish incorrect clinical labels. Labels remain unchanged.



Nested whole-group subsets use 25%, 50%, 75% and 100% of each training fold with unchanged validation rows. Error bars show fold standard deviation. Preprocessing is fitted inside each subset. The audit contains 90 fits, including reproduced baselines; no score at 20,000 rows is extrapolated.

| Full-size development condition         | Accuracy | Macro F1 | Emergency recall |
|-----------------------------------------|----------|----------|------------------|
| Logistic Regression / Prior features    | 86.09    | 86.57    | 92.65            |
| Logistic Regression / English details   | 87.40    | 87.83    | 93.38            |
| Logistic Regression / Original details  | 88.99    | 89.38    | 93.63            |
| HistGradientBoosting / Prior features   | 84.26    | 84.76    | 90.38            |
| HistGradientBoosting / English details  | 85.54    | 86.02    | 91.48            |
| HistGradientBoosting / Original details | 87.65    | 88.07    | 92.89            |
| Random Forest / Prior features          | 79.13    | 79.21    | 95.96            |
| Random Forest / English details         | 79.93    | 80.00    | 96.32            |
| Random Forest / Original details        | 80.24    | 80.32    | 95.89            |

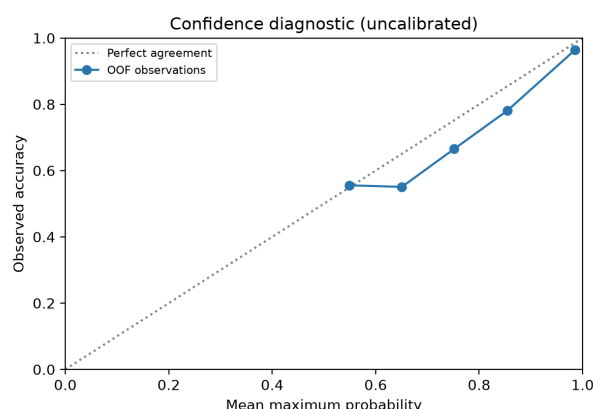
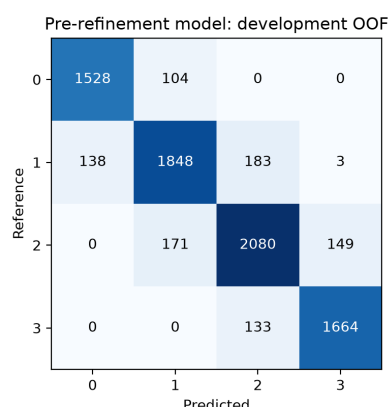
## Recommended next step

Review the local error queue and duration mismatches, agree a consistent four-level rubric with qualified reviewers, and collect distinct new examples. More rows may help, but duplicated templates or unclear labels will not reliably solve the remaining Urgent/Standard errors. Reserve new independently reviewed records for confirmation before further selection.

The GUI retains the original complaint for the 19 detail features and uses checked English for SapBERT. Similarity and cluster views use full 768-D vectors; stop-word removal is inactive. The report measures supplied concepts plus original details, not end-to-end live translation accuracy.

# Further Four-Level Error Investigation

This diagnostic uses only the 8,001 development records and the pre-refinement model. Supplied labels 0, 1, 2 and 3 remain unchanged. It is a diagnostic of model errors, not an independent assessment of clinical label correctness.



| Pre-refinement development diagnostic          | Value  |
|------------------------------------------------|--------|
| Out-of-fold errors                             | 881    |
| Errors between neighbouring levels             | 878    |
| Urgent / Standard confusions                   | 354    |
| Errors with confidence at least 90%            | 220    |
| All three classifiers agree on the wrong level | 345    |
| Matthews correlation coefficient               | 0.8522 |
| Multiclass log loss                            | 0.2648 |
| Multiclass Brier score                         | 0.1586 |
| Confidence calibration gap (10 bins)           | 0.0331 |

Higher MCC is better; lower log loss, Brier score and calibration gap are better. Confidence bins compare the mean maximum predicted probability with observed accuracy. These conditional development diagnostics do not measure end-to-end translation accuracy.

The follow-up freezes 13 settings before evaluation: LR regularization and balancing at PCA-128, PCA-64/256 alternatives, HGB leaf/regularization settings and RF feature sampling. All retain the 19 original-complaint details. Every setting uses five grouped folds; the original emergency-recall selection constraint remains fixed.

Selected development diagnostics: MCC 0.8561; log loss 0.2459; Brier score 0.1523; calibration gap 0.0120.

The older three-level scores concern a different labelling task. No labels are replaced, no difficult records are dropped and no test-based selection is used.

Latest refinement versus the immediately prior model: pooled development F1 gain +0.26 percentage points; conditional paired group interval -0.08 to +0.64. The interval includes zero, so a reliable gain is not established. Repeated-selection uncertainty is not included.